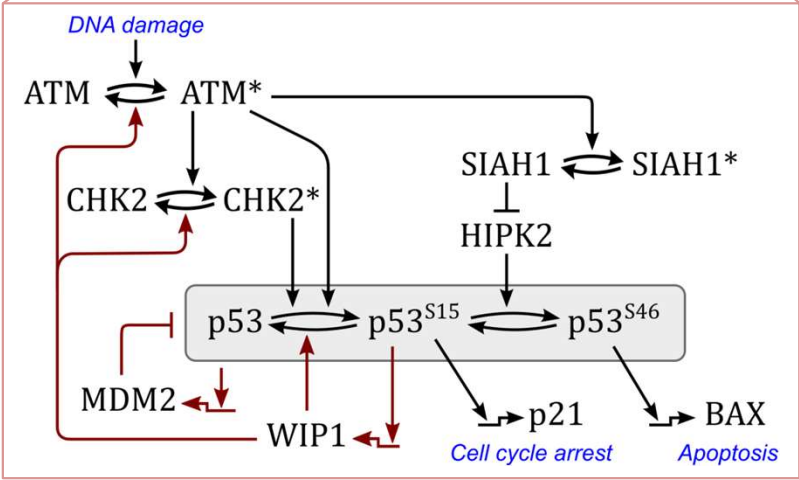
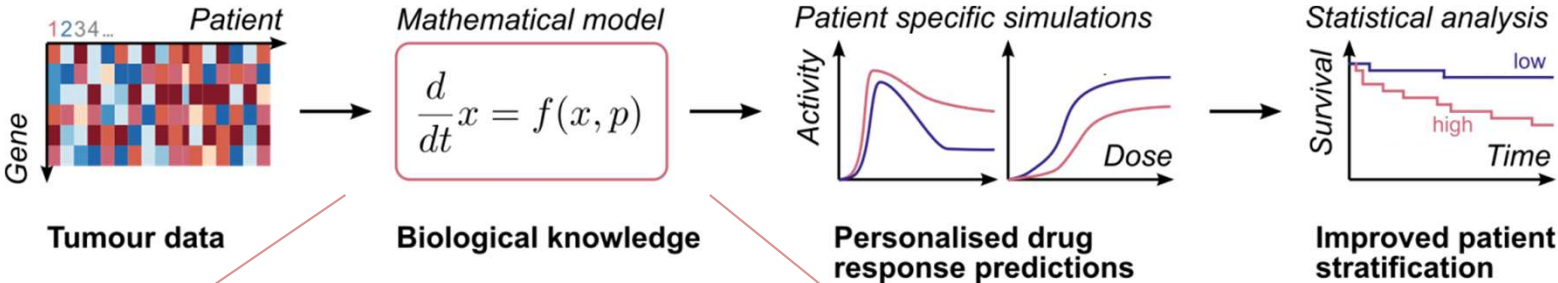


Modelling Project: Personalised modelling of the DNA damage response

Please select any cancer type (except neuroblastoma) treated with chemotherapy. Questions to address in the project (at least 2 should be answered):

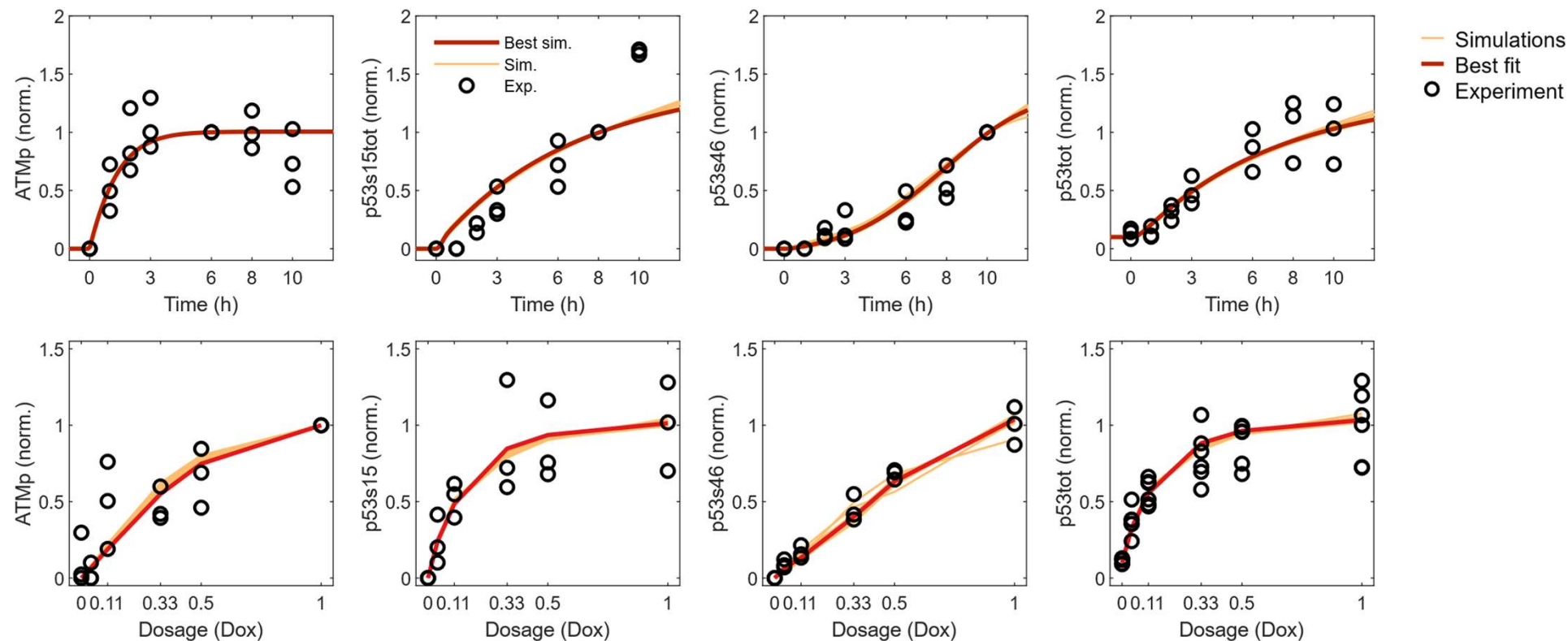
- 1. Can the p53 or any other ODE model predict patient survival (e.g. TCGA data)?**
- 2. Can the p53 or any other ODE model predict the responses to DNA damaging drugs in cell lines (use DepMAP & GDSC or PRISM data)?**
- 3. Can regression based model from cell lines predict patient survival?**
- 4. Does Cox regression based model from patient data correlate with sensitivity of cell lines towards chemotherapeutic drugs?**
- 5. How do the predictions of the mechanistic p53 of any other ODE model compare to ML-based approaches? => make the same predictions using (different) ML techniques such as regression or neuronal networks (deep learning) => You can use neuroblastoma for this task.**
- 6.* Can the p53 model predict DNA-damage induced gene expression signatures (use LINCS data)? This might not be possible. The first issue to figure out is whether there is a doxorubicin induced gene-signature apparent in the LINCS data.**

The existing p53 model



We can understand
how cancer cells compute
on the level of individual patients!

The model **simulates** experimentally measured time and dose response data.

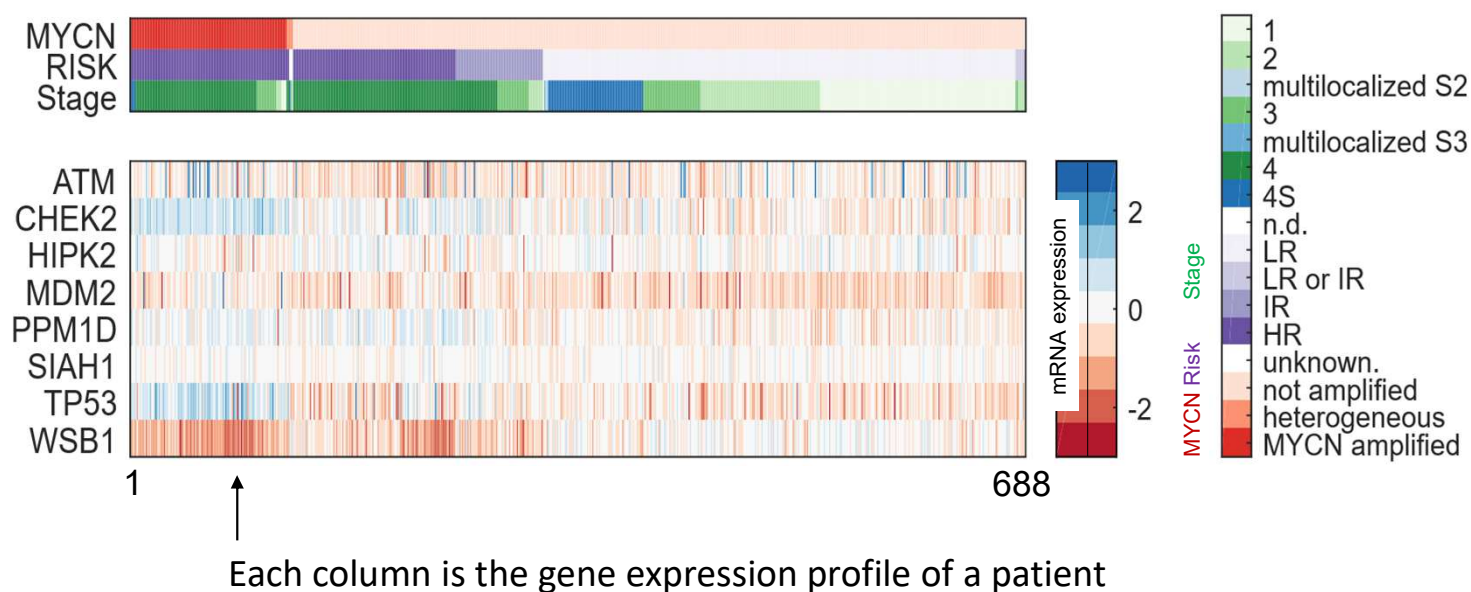


Experimental time-course data from

- [1] Our own experiments, neuroblastoma cell line SH-SY5Y
- [2] Batchelor et al. 2008 Recurrent initiation... Molecular Cell 30(3):277–289.
- [3] Purvis et al. (2012) p53 dynamics control cell fate. Science 336(6087):1440–1444

We used gene-expression data from neuroblastoma patients to personalise the p53 model

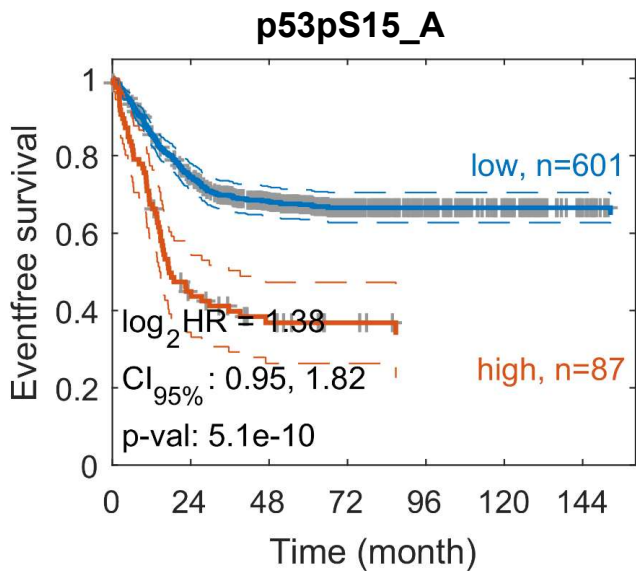
Training cohort: 688 patients, multi-centre, multi-national



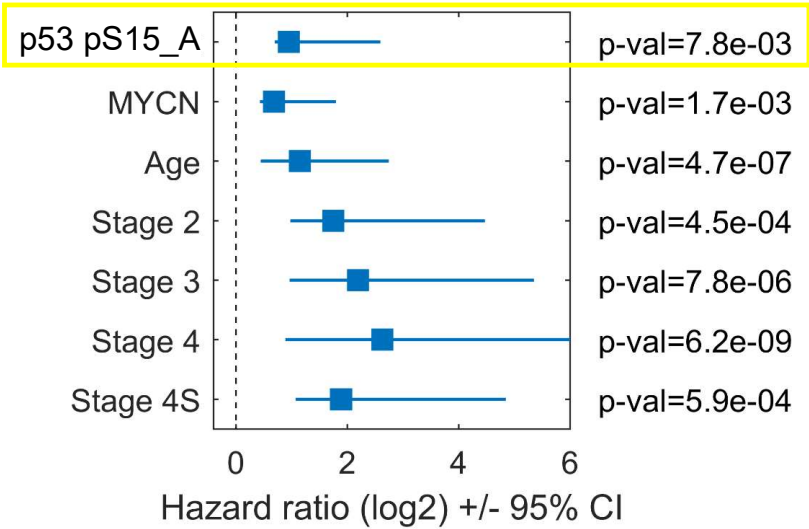
Neuroblastoma patient data from *Schramm et al 2015, Nature Genetics 47(8):892-877*

Model simulations **were prognostic** of event-free patient survival

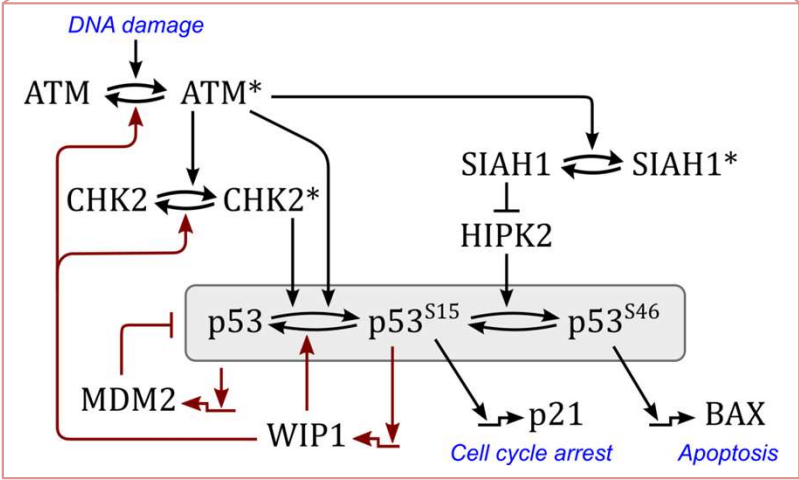
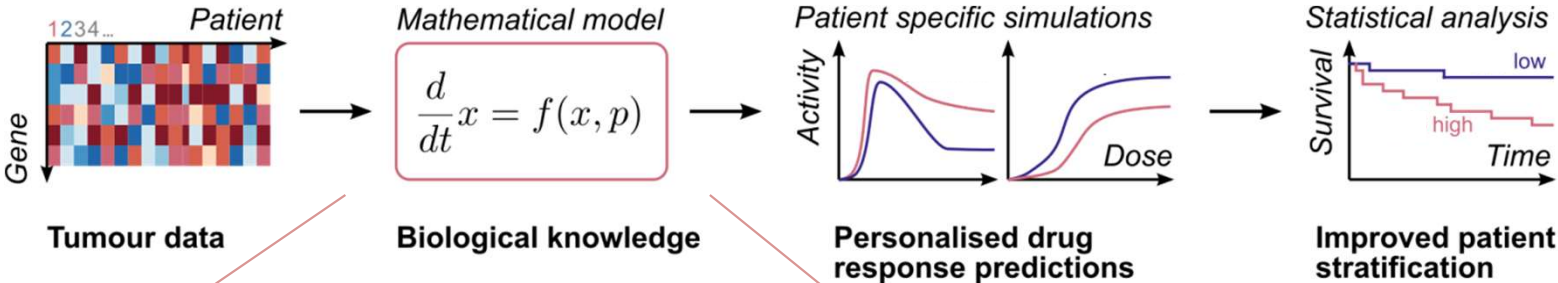
Training cohort: 688 patients



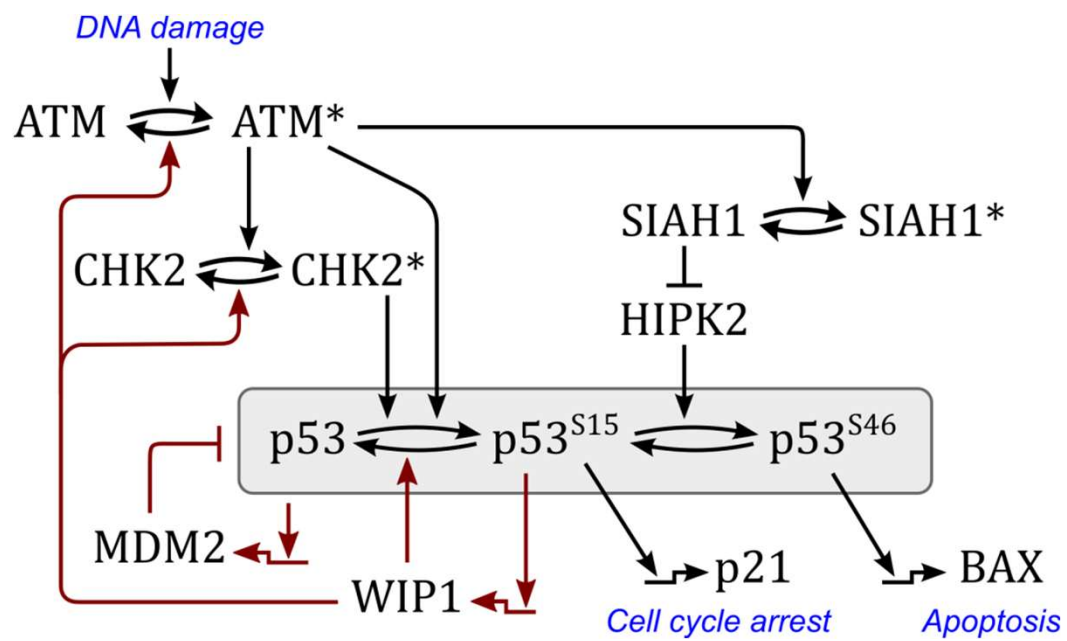
Model simulations **remain prognostic** after correction of confounding factors using Cox regression

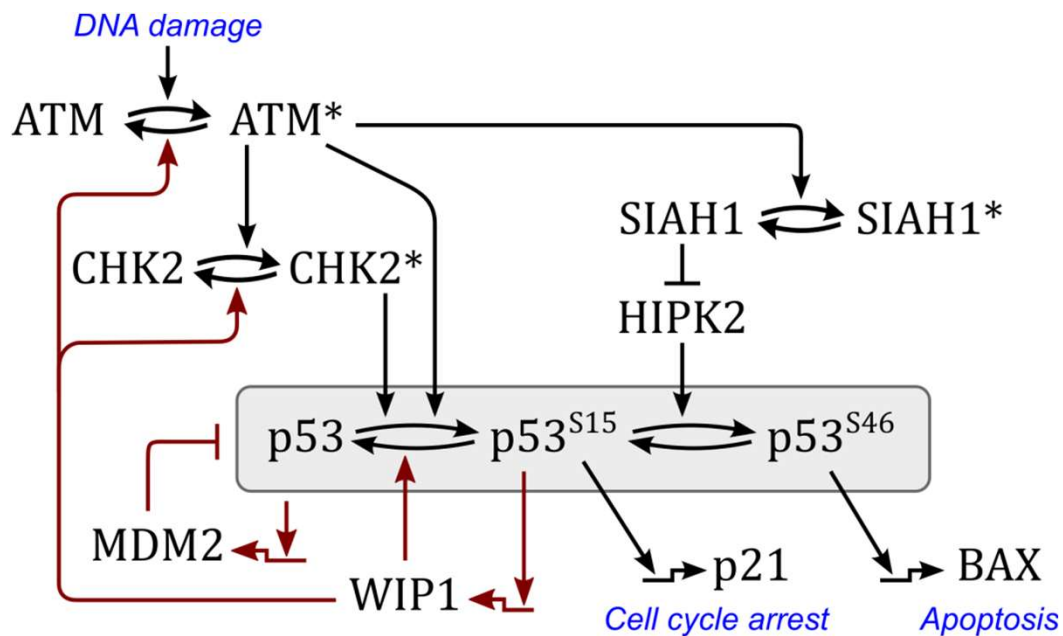


The existing p53 model



*We can understand
how cancer cells compute
on the level of individual patients!*





In the absence of DNA damage (=0):

⇒ Phosphorylation reactions become zero

Steady state (for DNA damage = 0):

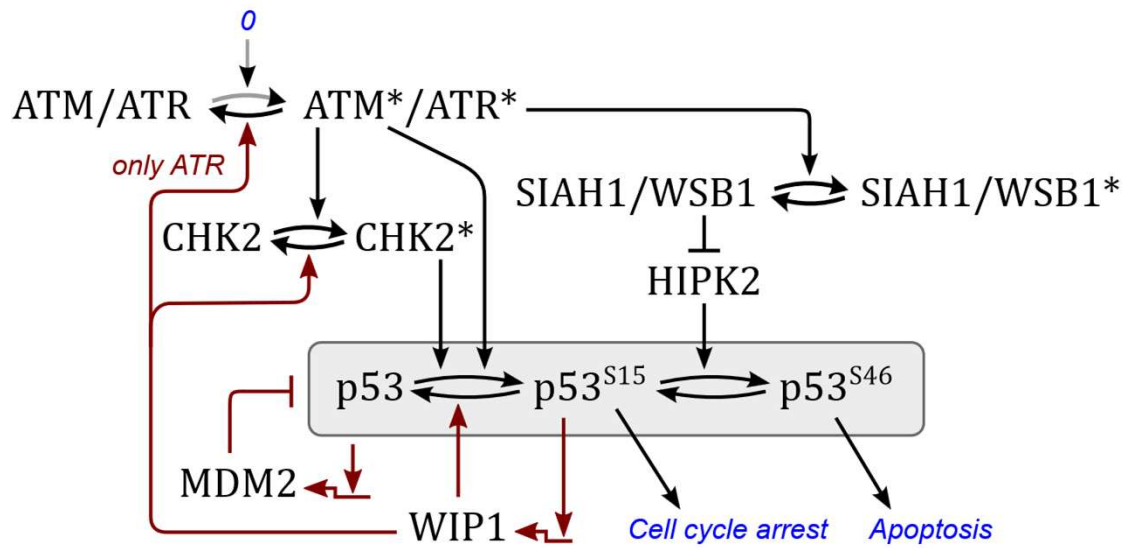
⇒ p53 is not phosphorylated:

$p53^{S15} = 0$, $p53^{S46} = 0$

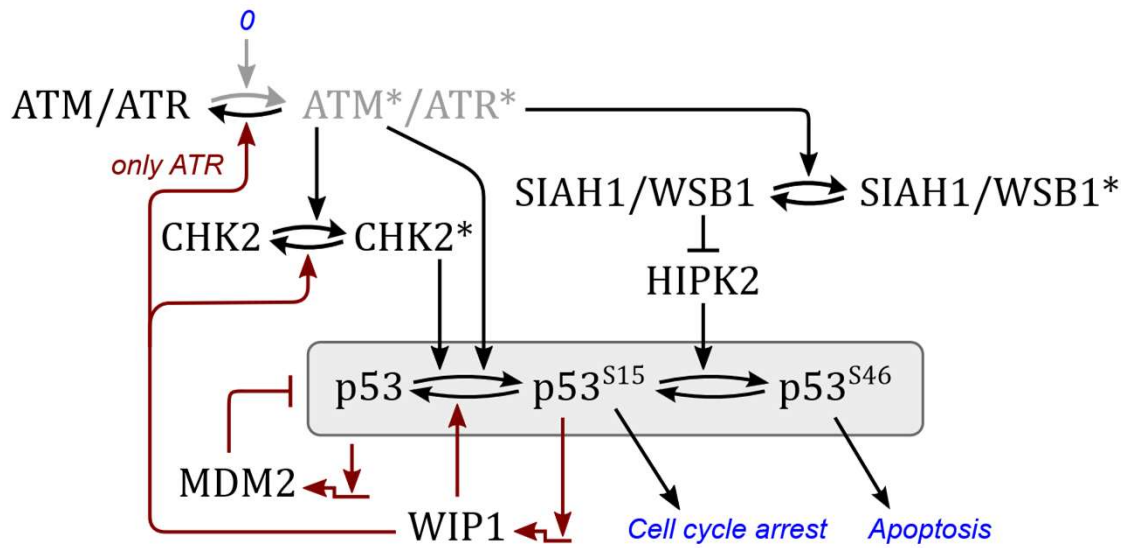
⇒ The model becomes much simpler

⇒ The patient-specific parameters have a “simple” formula

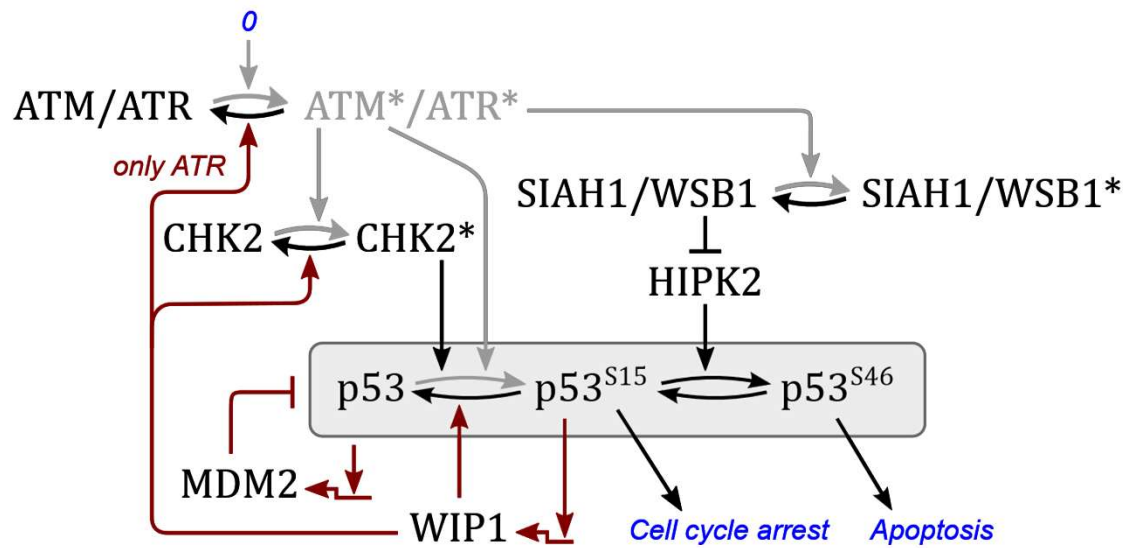
(can be solved analytically)



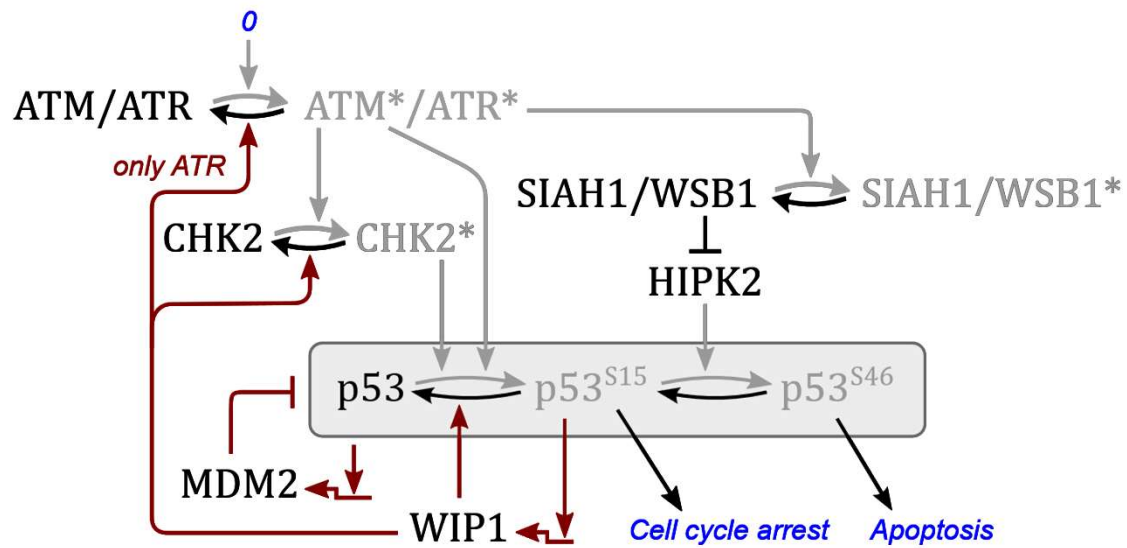
In the absence of DNA damage (=0):
⇒ Phosphorylation reactions become zero



In the absence of DNA damage (=0):
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 ⇒ Phosphorylation reactions become zero



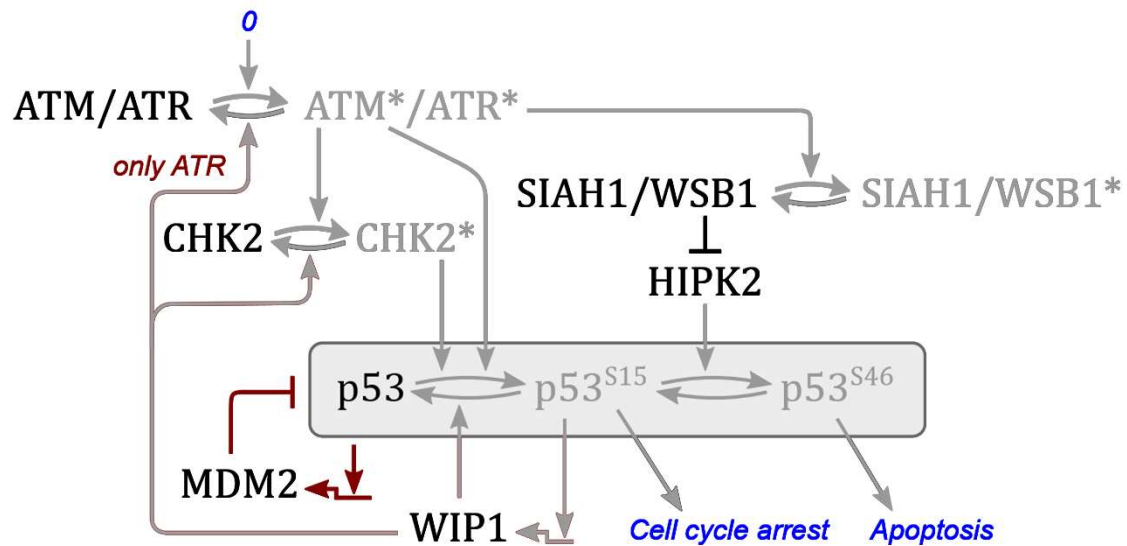
In the absence of DNA damage ($=0$):

$\Rightarrow$ Phosphorylation reactions become zero

Steady state (for DNA damage $= 0$):

$\Rightarrow$ p53 is not phosphorylated:

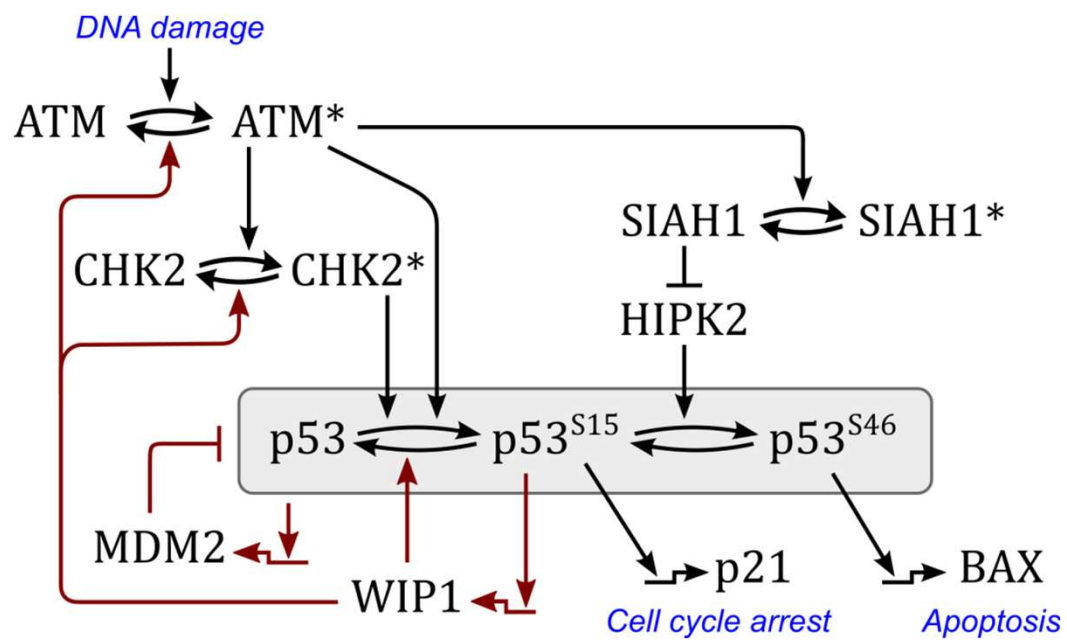
$p53^{S15} = 0, p53^{S46} = 0$



In the absence of DNA damage (=0):
 $\Rightarrow$ Phosphorylation reactions become zero

Steady state (for DNA damage = 0):
 $\Rightarrow$ p53 is not phosphorylated:
 $p53^{S15} = 0, p53^{S46} = 0$

$\Rightarrow$ The model becomes much simpler
 $\Rightarrow$ The patient-specific parameters have a
 "simple" formula
 (can be solved analytically)



How to get the data?

[DepMap Data Downloads](#)

 **depmap portal**

Enter a gene, cell line, lineage or compound

Tools ▾ Downloads About ▾ Forum

Data Downloads

TOP DOWNLOADS

CUSTOM DOWNLOADS

ALL DOWNLOADS

Type ▾

- ☐ Cellular Models
- ☐ Genetic Dependency
- ☒ Drug Sensitivity
- ☐ Other

DepMap Releases (CRISPR + Omics) ▾

- ☒ DepMap Public 21Q2
- ☐ DepMap Public 21Q1
- ☐ DepMap Public 20Q4 v2

Release	File	Date	Size	
Search in Release	☐ Search in File			
Sanger GDSC1 and GDSC2	sanger-dose-response.csv	12/19	91.06 MB	Download
Sanger GDSC1 and GDSC2	sanger-viability.csv	12/19	312.6 MB	Download

sanger-viability.csv

Download: [Download](#)

Source: Wellcome Trust Sanger Institute

File Description

Fold-change viability data for each cell line, drug, dose combination.

Sanger

Review paper:



Briefings in Bioinformatics, 22(1), 2021, 360–379

doi: [10.1093/bib/bbz171](https://doi.org/10.1093/bib/bbz171)

Advance Access Publication Date: 17 January 2020

Articles

Deep learning for drug response prediction in cancer

Delora Baptista, Pedro G. Ferreira and Miguel Rocha

Data resources: Drug responses, single drug

Data set	Website	# Compounds	# Cell lines	PHARM	MUT	CNV	mRNA EXP	miRNA EXP	METHYL	PROT	METAB
NCI-60[58]	https://dtp.cancer.gov/discovery_development/nci-60/	~50 000	60	x	x	x	x	x	x	x	x
GlaxoSmithKline [59]		19	311	x							
CCLE [1, 60, 61]	https://portals.broadinstitute.org/ccle	24	479*	x	x	x	x	x	x	x	x
GDSC1 [2]	https://www.cancerrxgene.org/	320	988	x	x	x	x		x		
GDSC2 [2]	https://www.cancerrxgene.org/	175	810	x	x	x	x		x		
CTRPv1 [3]	https://portals.broadinstitute.org/ctrp.v1/	354	242	x							
CTRPv2 [4]	https://portals.broadinstitute.org/ctrp/	481	860	x							
gCSI [62]	http://research-pub.gene.com/gCSI-cellline-data/	16	410	x	x	x	x				
FIMM [63]		52	50	x							

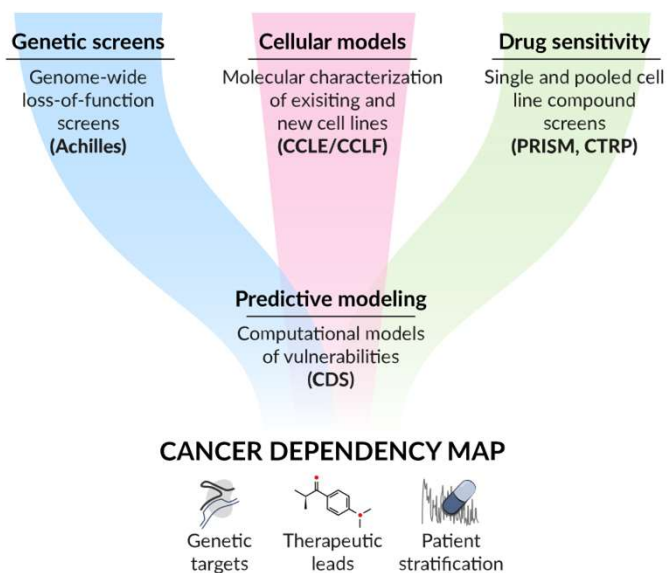
Data resources: Drug responses, drug combinations

Data set	Website	# Compounds	# Combinations	# Cell lines	PHARM	MUT	CNV	mRNA EXP	miRNA EXP	METHYL	PROT	METAB
Merck Compound Screen [64]		38	583	39	x							
NCI-ALMANAC [65]	https://dtp.cancer.gov/ncialmanac	104	5232	60	x	x	x	x	x	x	x	x
AstraZeneca-Sanger DREAM [8]	https://www.synapse.org/\#!Synapse:syn4231880	118	910	85	x							

Data resources: Genomic perturbations & responses

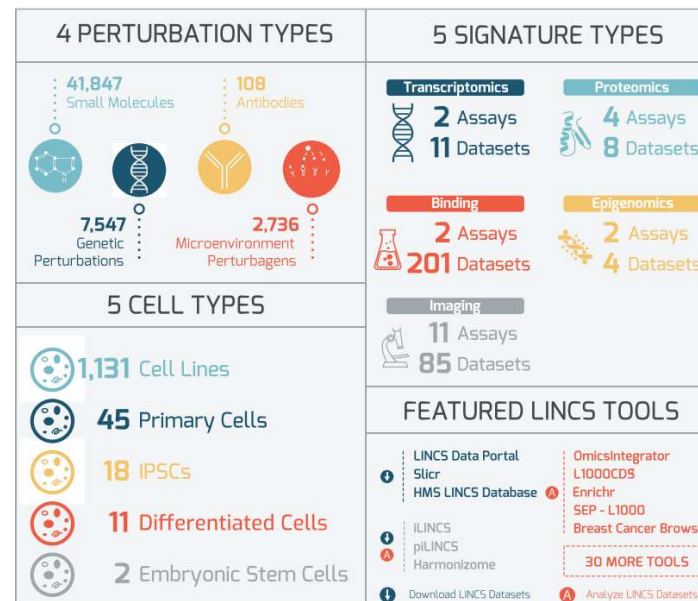
[DepMap: The Cancer Dependency Map Project at Broad Institute](https://depmap.org/portal/)

<https://depmap.org/portal/>



[LINCS: Library of Integrated Network-Based Cellular Signatures at NHI](https://lincsproject.org/LINCS/)

<https://lincsproject.org/LINCS/>

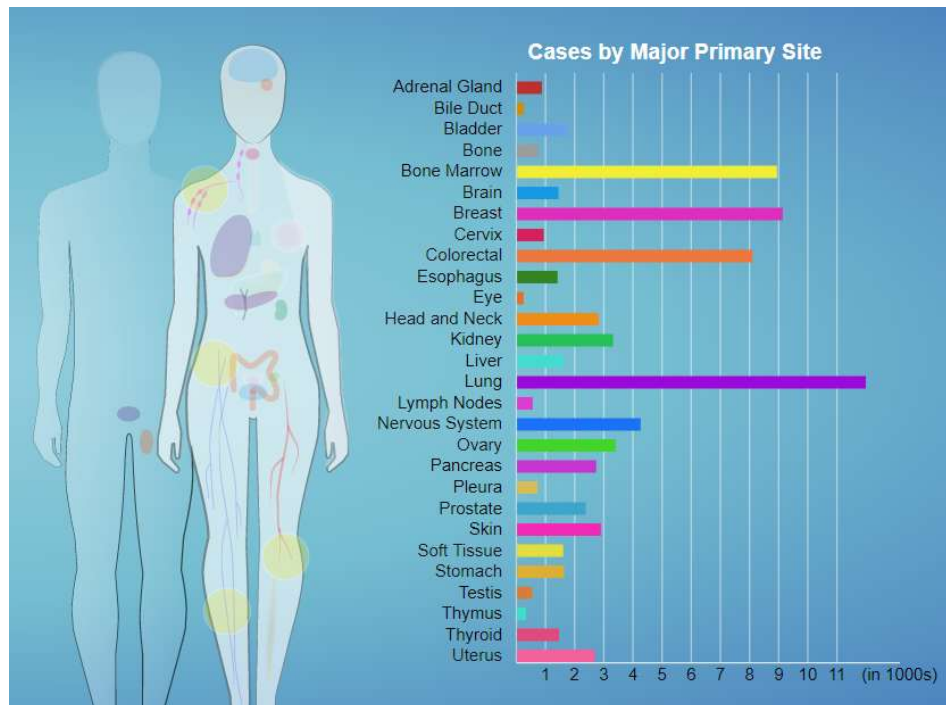


Cell Syst. 2018; 6(1):13-24. doi:10.1016/j.cels.2017.11.001

Data resources: Patent level genomic data (cancer)

[NCI Genomic Data Commons \(GDC\)](https://gdc.cancer.gov/)

<https://gdc.cancer.gov/>



[cBioPortal for Cancer Genomics](https://www.cbioportal.org/)

<https://www.cbioportal.org/>

Data sources

PubMed

NIH THE CANCER GENOME ATLAS
National Cancer Institute
National Human Genome Research Institute

International Cancer Genome Consortium

Count Me In

CCLE Cancer Cell Line Encyclopedia

AACR American Association for Cancer Research
FINDING CURES TOGETHER™

PROJECTGENIE
Genomics Evidence Neoplasia Information Exchange